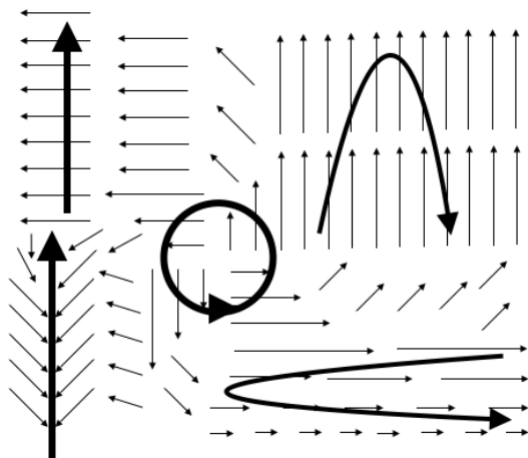


Math 2551 Worksheet: Vector Line Integrals

1. Consider the vector field $\mathbf{F}$ (thin arrows) and let $\mathbf{T}$ denote the unit tangent vector to the directed curves shown below (denoted with thick arrows). Determine whether

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds$$

is positive, negative, or zero for each directed curve C . In other words, determine whether the *work* done by the vector field on each curve is positive, negative, or zero.



2. Evaluate $\int_C (2x - y)\mathbf{i} \cdot d\mathbf{r}$ where C is parameterized by $\mathbf{r}(t) = (t^2)\mathbf{i} + (3t - 2)\mathbf{j}$, $t \in [0, 1]$.
3. Find the work done by the force $\mathbf{F} = xy\mathbf{i} + (y - x)\mathbf{j}$ over the straight line from $(1, 1)$ to $(2, 3)$.
4. Consider the closed curve C consisting of a semicircle and a straight line segment as follow:

$$\mathbf{r}_1(t) = (2 \cos t)\mathbf{i} + (2 \sin t)\mathbf{j}, \quad t \in [0, \pi], \quad \mathbf{r}_2(t) = t\mathbf{i}, \quad t \in [-2, 2]$$

Let the vector field $\mathbf{F}$ be given by

$$\mathbf{F}(x, y) = -y^2\mathbf{i} + x^2\mathbf{j}.$$

Find the circulation of $\mathbf{F}$ around C and the flux of $\mathbf{F}$ across C .

5. Give an example of a non-trivial force field $\mathbf{F}$ (not the zero vector at all points) and a non-trivial path $\mathbf{r}(t)$ (not the stationary path at a point P) for which the total work done moving along the path is zero.